

VISWESWAR SIRISH PARUPUDI

New York, NY | +1 (475) 348-4238 | vsp7230@nyu.edu | [linkedin.com/in/sirish-visweswar](https://www.linkedin.com/in/sirish-visweswar) | github.com/itsSirish | itssirish.github.io

EDUCATION

New York University | *Master of Science, Computer Science*

May 2026

GPA: 3.85/4.0 | Coursework: Reinforcement Learning, Machine Learning, Software Engineering, Big Data, Algorithms, Game Design

Birla Institute of Technology and Science, Pilani | *Bachelor of Engineering, Electronics and Communication*

May 2024

Minor in Computing and Intelligence

EXPERIENCE

Research Engineer | **Neuroinformatics Lab, NYU**

New York, NY | June 2025 – Present

- Adapted and designed audio-transformer-based foundation models for real-time in-vivo brain region localization on **5 TB** of raw multichannel Neuropixels time-series spanning the full mouse brain atlas with novel electrode-geometry-aware self-supervised contrastive objectives.
- Achieved state-of-the-art 3D coordinate regression at **203 μm** RMSE across the 1 cm mouse brain (**75% lower error** than the previous best), outperforming supervised ViT, CNN, and gradient-boosting baselines with zero-shot generalization across probes, labs, and species.
- Scaled **95M-param** training via multi-GPU DDP and Ray Tune (HEBO/ASHA) on H200/L40S SLURM, with weights on Hugging Face.
- Owned the acquisition-to-reconstruction pipeline for a high-speed two-photon microscope recording cell membrane voltage at 440 Hz to capture spike bursts, resampling its non-uniform sinusoidal scan trajectory onto a regular grid via an adaptive-bandwidth Gaussian kernel.
- Designed a **self-supervised video denoiser** trained without clean ground truth (blind-spot training), pairing a factorized space-time transformer with a CNN and sparsity penalty, separating action potentials from shot noise; **reaching 98% noise suppression at 100% signal retention**.
- Developing cross-modal 3D image correspondence for in-vivo/ex-vivo cell matching using **fine-tuned DINOv2 features**, attention across volume slices, and landmark supervision, generalizing to held-out subjects.

Graduate Assistant | **NYU** | **Computer Vision, Neuroinformatics**

New York, NY | Aug. 2025 – May 2026

- Ran live coding sessions, office hours, and research project mentorship for **100+** graduate students, and **curated 8 Kaggle competitions** ([Link](#)) end-to-end (dataset curation, baselines, metrics, and submission evaluation) across advanced CV and Neuroscience domains.

Data Science Research Intern | **Speech & Language Lab, NTU Singapore** | [Link](#)

Remote | June 2023 – Dec. 2023

- Developed an information extraction pipeline (undergraduate thesis, [Link](#)) on noisy social-media text targeting downstream sentiment analysis and hate-speech detection, using LLM-based synthetic data generation (seq2seq paraphrasing, few-shot Llama 2 via LangChain) with BERTScore quality filtering, improving extraction performance by **8.6% F1** with a **24% fluency gain**.

Data Science Intern | **DLT Labs (KNNX)** | [Certificate](#)

Hyderabad, India | May 2022 – July 2022

- Built a real-time Ethereum gas-free forecasting service with minute-level ingestion; reduced **prediction error by 27%** via time-series analysis, ARIMA selection, ACF/PACF diagnostics, and grid-search hyperparameter tuning, with data-quality alerting ensuring **99%+ uptime**.

PUBLICATIONS

[**NeurIPS 2026, CORE A***] Geometry-Aware Self-Supervised Electrophysiology Representation Learning | *Co-first author*

[**NeurIPS 2026 BrainBodyFM Workshop**] LFP2Gene: Recovering Local Molecular Composition from Local Field Potentials | *Co-first author*

[**Nature Methods, under review**] OJAMI: Electro-Optic Jitter Accelerated Multiphoton Imaging | *Co-author*

[**IEEE Sensors Letters 2023**] A Smartphone-Enabled Deep Learning Approach for Myocardial Infarction Detection Using ECG Traces for IoT-Based Healthcare Applications | *First author*

SELECTED PROJECTS

CineMatch – Group Movie Matching & Social Discovery Platform | *React, Django, PostgreSQL, Redis, AWS, LLMs* | [Link](#)

- Shipped a full-stack collaborative movie discovery platform with a normalized relational schema, RESTful microservices, role-based group permissions, and **WebSocket-backed** real-time chat via Redis pub/sub for **low-latency** group matching and synchronized swiping.
- Led** a 6-person agile team across 8 sprints; integrated an **Agentic recommender** (Llama 3.3 70B via Groq) with catalog search, watchlist retrieval, and group-preference aggregation, generating ranked group recommendations with natural-language explanations over a **50K+ title catalog**.
- Established CI/CD via **Travis CI** with **285 automated tests**, 90%+ coverage and load testing under concurrent multi-user sessions.

Det-IGEN: Detectable Watermarking in AI Image Generation | *Python, PyTorch, GenAI, Docker* | [Link](#)

- Implemented a training-free, invisible **Tree-Ring watermarking** layer in **Stable Diffusion 2.1** for AI-content provenance, embedding a key (zero/Gaussian/ring patterns) into the noise latent's Fourier spectrum, robust against image transformations with negligible quality impact.
- Integrated a detector that recovers the initial noise via **DDIM inversion** and matches the key by Fourier-domain distance, separating watermarked images from unmarked and real COCO photos at **99.5% accuracy** and **100% recall** (zero missed AI-generated images).

TECHNICAL SKILLS

Languages: Python, C/C++, Java, JavaScript/TypeScript, SQL, MATLAB, HTML/CSS | **Web:** React, Django, Express/Node.js, REST APIs

ML/AI: PyTorch, TensorFlow, Keras, Hugging Face Transformers, scikit-learn, LangChain, LLMs, NumPy, Pandas, OpenAI Gym, Unity

Infra & Cloud: AWS (EC2, Elastic Beanstalk, RDS, S3), Docker, Kubernetes, Kafka, Spark, Redis, PostgreSQL, SLURM/HPC, Git, CI/CD